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A longitudinal study on the lasting impact of Pandemic-induced online learning

Long Ma[1] · **Chei Sian Lee**[2]

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Abstract
Although the COVID-19 pandemic has subsided, lingering negative experiences with online learning continue to exert significant long-term influences. The pandemic serves as a disruptive intervention that warrants an in-depth investigation to understand how such crises influence long-term perceptions and online learning practices. Using the Technology-User-Environment theoretical framework, this longitudinal study examines the key factors influencing students' intentions to engage in online learning, as well as how the effects of these factors have evolved before, during, and after the pandemic. A total of 2,250 valid questionnaires were collected through three waves of independently pooled cross-sectional data spanning from 2016 to 2024. The data was analyzed using ANOVA and multiple regression techniques. Key findings indicate students' intentions of engaging in online learning declined significantly during the pandemic but showed a notable rebound post-pandemic. Specifically, perceived benefits such as usefulness, performance-to-cost value and accessibility, exhibited a fluctuating impact on online learning intentions, with a sharp decline during the pandemic followed by a rebound in the post-pandemic period. The importance of self-regulation skills was amplified during the pandemic, underscoring the necessity of strong self-regulatory capabilities in remote learning environments. In contrast, the negative influences of traditional learning preferences and inadequate institutional support were significantly intensified during the pandemic and continued to hinder online learning intention after the pandemic. These insights deepen our understanding of the dynamic impacts of the key factors on online learning, and offer valuable implications for educational institutions and policymakers to address and mitigate the adverse effects of the pandemic on individuals' online learning intentions.

Keywords Online learning · COVID-19 pandemic · Lasting effects · Continuous usage intention · MOOCs · TUE framework · Longitudinal study

Extended author information available on the last page of the article

1 Introduction

The global COVID-19 pandemic (henceforth, the pandemic) has caused unprecedented disruption to the education systems worldwide, impacting approximately 1.6 billion learners across more than 200 countries and bringing multilevel effects to the education ecosystem (Reimers, 2024; UNESCO, 2020). With lockdown and social distancing policies, educational institutions were compelled to close campuses and transition to online teaching. This led to the widespread adoption of online learning on a large scale. While universities faced the challenge of adapting to online education during the pandemic, it triggered an unexpected digital transformation within the entire education system (DeVaney et al., 2020; Hussein et al., 2020). The extensive implementation of online learning has prompted the belief that it will reshape the landscape of higher education, evolving into a regular and essential component of pedagogical methods after the pandemic (Mali & Lim, 2021; Szopiński & Bachnik, 2022). Online learning is increasingly recognized as a sustainable and future-ready educational model, owing to its accessibility, adaptability, affordability, and resource diversity (Adedoyin & Soykan, 2023; Kim et al., 2023; Ratten, 2023). Put simply, online learning is essential in an increasingly digital world.

Contrary to optimistic expectations, however, studies focusing on learners' attitudes and motivations toward online learning during the pandemic have reported contrasting findings (Shirish et al., 2021; Toney et al., 2021). A broad body of research confirms that students found learning during the pandemic unsatisfactory, with many expressing negative views of online learning (Aguilera-Hermida, 2020; Usher et al., 2021). Several factors contributed to students' dissatisfaction with online learning during the pandemic, including distractions, difficulties in concentration, increased workload, technological problems, and insufficient support (Maqableh & Alia, 2021). Importantly, such negative experiences were particularly prominent among learners in developing countries, where challenges such as lack of internet access, limited access to digital devices, and financial constraints emerged as significant barriers to online learning during the pandemic (Adnan, 2020; Bestiantono et al., 2020). While students in developed countries benefited from stronger infrastructure, they also experienced substantial challenges such as screen fatigue, low engagement, mental health struggles, and reduced interpersonal connection with peers and instructors during the pandemic (Alon et al., 2023; Gherghel et al., 2023). In summary, although the challenges differed in form and magnitude, research has demonstrated that the pandemic had a profound and widespread impact on online learning worldwide.

Notably, one of the most significant challenges during the pandemic was sustaining students' motivation and engagement in online learning (Chiu, 2021; Salas-Pilco et al., 2022; Teodorescu et al., 2022). While some students were reportedly able to adjust to the online learning pedagogy, the majority found it challenging (Pokhrel & Chhetri, 2021). This lack of motivation prompted some students to consider taking leaves of absence or withdrawing from their studies (Gherghel et al., 2023). The negative experiences associated with online learning during the pandemic may lead to a range of biased attitudes toward online learning activities, potentially deterring students from embracing online learning in the future (Baber, 2021; Lemay et al., 2021).

Despite the negative attitudes and experiences during the pandemic, online learning offers advantages such as flexibility, convenience, diverse course offerings, cost-effectiveness, and opportunities for career advancement, making it a valuable option for individuals seeking accessible and personalized education (Hussein et al., 2020; Szopiński & Bachnik, 2022). Therefore, the pandemic represents a significant intervention that merits in-depth examination to understand how it has shaped perceptions and practices in online education. The examination can provide insights into leveraging the strengths of online learning while addressing its challenges, ultimately fostering a more effective digital learning landscape. This understanding is essential for rekindling and sustaining students' learning interests and motivations (Fang et al., 2023). Consequently, the present study seeks to investigate how the learning experiences during the pandemic have shaped students' intentions to engage in online learning in the post-pandemic era.

Importantly, the pandemic unfolded during these students' formative years, leaving lasting impacts on their educational experiences and making this study especially relevant (Navarro et al., 2024; Wang et al., 2022). While previous studies have compared students' online learning experiences, motivations, and attitudes toward online learning before and during the pandemic (e.g., Caron et al., 2022; Hicks et al., 2021; Lin, 2021; Maqableh & Alia, 2021), few have longitudinally examined the evolution of these effects and their enduring impacts in the post-pandemic era. Therefore, the objective of the present study is to investigate the factors that have influenced students' intentions to use online learning during the pandemic and how the effects of these factors have evolved before, during, and after the pandemic. Utilizing the Technology-User-Environment (TUE) framework, the study explores influential factors derived from technologies (perceived usefulness, performance-to-cost value, accessibility, lack of interactivity), users or students (self-regulation, usage experience), and social environments (learning tradition, peers' impact, lack of instruction, and lack of publicity) (Gupta, 2020; Khalifa et al., 2024; Ma & Lee, 2019). Through three phases of data collection among college students, this study also examined the dynamic effects of various factors across the pre-, during, and post-pandemic phases. The findings have the potential to impact online learning practices, inform policy decisions, and contribute to a more effective and inclusive online learning landscape in the post-pandemic era while enhancing preparedness for future disruptions. Accordingly, this study aims to explore the following research question: *How have disruptions such as the pandemic changed students' attitudes and learning intentions toward online learning over time*?

2 Literature review

2.1 Learning experiences during the pandemic

A number of studies have extensively explored the advantages and challenges encountered by students during online learning during the pandemic (Adedoyin & Soykan, 2023; Maatuk et al., 2022; Nicholson et al., 2023; Gopal et al., 2021). Among the advantages, past research emphasized aspects such as cost- and time-effectiveness,

flexibility, safety, convenience, and the enhancement of digital competencies. (Hussein et al., 2020; Alon et al., 2023). In contrast, a predominant body of prior research has consistently highlighted the negative impacts of online learning experiences during the pandemic on students' motivations, perceptions, well-being, and attitudes (Daniels et al., 2021; Lemay et al., 2021). For example, Hussein et al. (2020) identified adverse effects, including distraction and reduced focus, a heavy workload, technological and internet problems, and insufficient support from instructors. Baltà-Salvador et al. (2021) found students were dissatisfied with the quality of online education and reported increased feelings of discouragement, boredom, and confusion. Similarly, Gherghel et al. (2023) noted that the abrupt shift to online learning negatively impacted students' mental health, leading to demotivation and disengagement. Consequently, students tend to anticipate future online learning experiences negatively (Caron et al., 2022; Usher et al., 2021; Zhu et al., 2023). While numerous studies have focused on identifying the detrimental impacts of adverse experiences on students' learning motivations and attitudes, few have examined whether the adverse effects have persisted into the post-pandemic era. Specifically, it remains unclear whether students have sustained a diminished intention to engage in online learning due to the lasting effects of negative experiences or if they have recovered from these challenges and are embracing online learning. In summary, the pandemic's forced shift to digital platforms offers a unique opportunity to study how students' experiences have influenced their long-term attitudes and intentions toward online learning, serving as a significant intervention.

Notably, the studies that have compared students' online learning experiences before and during the pandemic have relied on students' remembered pre-pandemic experiences of online learning (Caron et al., 2022; Daniels et al., 2021; Hicks et al., 2021). Findings based on remembered experiences, often dating back more than a year, may introduce bias and fail to capture students' accurate perceptions and attitudes, which could be significantly influenced by their current experiences. Drawing on longitudinal data collected before, during, and after the pandemic, this study aims to investigate the factors that influence students' intentions to engage in online learning and to explore how the effects of these factors have evolved over time. Recognizing that students' perceptions of online learning can be influenced by various technological, institutional, contextual, and personal factors (Lemay et al., 2021), the study adopts the Technology-User-Environment theory as the theoretical framework to analyze the influential factors and their dynamic impacts on students' online learning intentions.

2.2 The Technology-User-Environment framework

In this study, we employed a context-specific approach to explore the influential factors derived from students' perceptions of online learning, namely Technology-User-Environment (TUE) framework (Baker, 2012; Gupta, 2020; Ma & Lee, 2019). Such a context-specific approach has been widely applied by previous studies (Kim et al., 2023; Shirish et al., 2021). The TUE framework classified the influential factors of online learning adoption into three main aspects: technology (i.e., perceived usefulness, performance-to-cost value, accessibility, and lack of interactivity), users (i.e.,

self-regulation and usage experience of online learning), and social environments (i.e., learning tradition, peers' impact, lack of instruction, and lack of publicity). The relationships between these factors and students' intentions to use online learning are articulated as follows.

2.2.1 Technological factors

Perceived usefulness Perceived usefulness is defined as the degree to which students believe that online learning would enhance their learning performance and effectiveness (Ma & Lee, 2019). Perceived usefulness has been found to be positively related to students' use of online learning tools during the pandemic (Baber, 2021; Yao et al., 2022). However, the negative learning experiences of online learning such as distractions, technical problems and increased workload during the pandemic might overshadow the perceived usefulness, potentially diminishing the positive impact on learning performance (Conrad et al., 2022; Maatuk et al., 2022). Consequently, these negative experiences may lead students to perceive online learning as less useful than initially anticipated, which in turn may further decrease their intentions to engage in online learning after the pandemic.

Performance-to-cost value Performance-to-cost value is a concept rooted in consumers' comparisons between the benefits and costs associated with innovations (Rogers, 2003). A study by Lin (2021) concluded that students generally acknowledged the overall benefits perceived in the online learning environment after the COVID-19 outbreak. However, prior studies also highlighted that students had to shoulder the added responsibility of mastering digital tools and adapting to new modes of learning (Adedoyin & Soykan, 2023). This added workload is anticipated to result in unforeseen financial and time costs which might lead to a reevaluation of the performance-to-cost value of online learning after the pandemic.

Accessibility Accessibility, encompassing internet access and digital device availability, emerged as a critical factor in students' online learning experiences during the pandemic (Adnan, 2020; Slack & Priestley, 2023). Prior research has indicated that accessibility features in online learning significantly influence students' engagement (Kim & Frick, 2011; Maheshwari, 2021; Salas-Pilco et al., 2022). However, during the pandemic, issues such as poor internet connectivity and lack of digital devices were identified as significant barriers to accessing online classes and learning resources, leading to dissatisfaction and reluctance to adopt online learning (Maqableh & Alia, 2021). In particular, students from remote areas and marginalized communities encountered more difficulties due to poor internet connectivity, the affordability of digital devices, and unfavorable study environments (Adnan, 2020).

Lack of Interactivity Interactivity, defined as direct communication and social interaction within online communities, is recognized as an essential motivator for active learning (Kop et al., 2011; Ma & Lee, 2019). During the pandemic, students experienced a decline in academic performance due to the lack of in-person consultations with instructors and the absence of peer interactions (Aguilera-Hermida, 2020;

Pokhrel & Chhetri, 2021). The absence of interaction with instructors and peers was identified as a key factor contributing to students' dissatisfaction with online learning (Selvaraj et al., 2021). Furthermore, the pandemic significantly limited opportunities for social interaction outside the classroom (Gherghel et al., 2023). Put differently, the absence of in-person interaction, coupled with reduced motivation stemming from limited interaction with instructors and peers, presents a major obstacle to students' willingness to adopt online learning after the pandemic.

2.2.2 Individual factors

Self-regulation Self-regulation, defined as individuals' ability to control goals, task strategies, time management, and environment structuring, has emerged as a crucial factor in students' online learning experiences during the pandemic (Hamdan et al., 2021; Han et al., 2023). In the online context, students faced challenges in maintaining focus and concentration, particularly in the home environment where distractions and a lack of proper study spaces were prevalent (Hussein et al., 2020; Maimaiti et al., 2023). In addition, the pandemic introduced additional obstacles, such as increases in stress, gaming, procrastination, and decreases in academic motivation (Göksu et al., 2021; Usher et al., 2021). Then, the importance of self-regulation skills became evident in overcoming challenges related to the issues above.

Usage experience Usage experience with online learning is a critical factor influencing students' perceptions and usage intentions (Lin, 2021). The swift transition to online learning during the pandemic, as noted by Daniels et al. (2021) and Oliveira et al. (2021), presented challenges for both students and instructors lacking relevant experience. Studies suggested that lower learning loss was associated with prior experience using technology, emphasizing the importance of pre-existing familiarity with technology-enhanced learning (Fütterer et al., 2023). Familiarity with technology emerged as a key component, encompassing the level of knowledge and experience with various forms of technology. Therefore, we argue that the students' experiences of online learning can impact their continuous usage intentions in the post-pandemic era.

2.2.3 Environmental factors

Learning tradition Learning tradition and habits play a crucial role in shaping students' perceptions and preferences toward online learning (Al-Adwan et al., 2022). Online learning often struggles with effective communication, hindered by low engagement, limited interactivity, perceived impersonal nature of computer-mediated interactions (Alawamleh et al., 2020). During the pandemic when challenges in communication, practical sections, and socialization were pronounced, the influence of learning traditions may further strengthen students' inclination towards face-to-face learning. The pandemic's negative impact on communication, practical engagement, and peer interaction likely reinforced students' traditional learning preferences, increasing their tendency to favor in-person learning.

Peers' impact Research emphasizes that peer support and interaction are crucial for creating a conducive online learning environment (Kop et al., 2011; Lee et al., 2017). Studies indicate that social support from peers and instant feedback from instructors contribute to increased satisfaction, engagement, and motivation among students (Gutiérrez-Santiuste et al., 2015; Tseng & Walsh, 2016). However, due to the social restrictions during the pandemic, students found it challenging to obtain effective peer support for engaging in online learning (Hasan & Khan, 2020). Lack of support from peers can hinder students' adoption of online learning platforms, as social interaction is integral to the learning experience (Ma & Lee, 2019). As a result, students may reevaluate the value of online learning and become reluctant to use online learning once face-to-face learning becomes available.

Lack of Instruction Students' perceived instructional support, including clear instructions, instructor feedback, and technical assistance, significantly influences their learning performance and engagement (Han & Geng, 2023). The abrupt transition to online instruction during the pandemic resulted in low-quality learning materials, limited IT support and training for end users, and the adoption of ineffective teaching strategies.(Shao et al., 2023; Shirish et al., 2021). As a result, both instructors and students faced challenges related to the abrupt transition, lack of preparation, and inadequate support systems, leading to a surface approach to teaching and learning and restricted use of technologies (Hasan & Khan, 2020; Simamora et al., 2020). Consequently, these negative learning experiences may decrease students' intentions to continue to utilize online learning after the pandemic.

Lack of Publicity In this study, publicity refers to the deliberate use of mass communication channels by educational institutions to actively promote and disseminate information about online learning among students (Loda & Coleman, 2005). The strategic promotion of online learning is vital in informing students about the availability of innovative resources in the online learning landscape (Ma & Lee, 2019). Throughout the pandemic, instructors and students encountered numerous technological and instructional challenges with online learning (Adedoyin & Soykan, 2023; Maatuk et al., 2022). However, the absence of adequate institutional preparedness meant that few effective strategies were in place to mitigate the emerging challenges.

In sum, through the lens of the TUE framework, we highlight how technological, individual, and environmental factors may shape students' online learning intentions across different phases of the pandemic. While prior research has explored these factors in isolated time frames, there is a lack of longitudinal studies capturing their evolving influence. To address this gap, the present study employs a three-wave cross-sectional survey design to examine how students' online learning intentions have been affected before, during, and after the pandemic.

3 Methodology

3.1 Data collection

To address the research question, the questionnaire survey method was employed to collect data, and the constructs were measured using adapted items from previous studies (Lian & Yen, 2014; Ma & Lee, 2019; Novak & Hoffman, 1997; Tangney et al., 2004; Venkatesh et al., 2003). In all, twenty-eight question items were utilized to measure the ten constructs (i.e., perceived usefulness, performance-to-cost value, accessibility, lack of interactivity, self-regulation, learning tradition, peers' impact, lack of instruction, lack of publicity, and intention to use) (in Appendix Table 6). Responses were recorded on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). The respondents' usage experience of online learning was measured by asking "how many times do you conduct online learning each week recently?". It was treated as a control variable in the analysis, consistent with previous studies (e.g., Ma & Lee, 2019; Wang et al., 2013).

We operationalized the online learning platform in the context of Massive Open Online Courses (MOOCs). MOOCs were employed as the operational context for the online learning platform to provide a consistent and standardized basis for comparison across different institutions and study periods. Their open-access nature, global reach, and uniform delivery format make MOOCs a suitable control for longitudinal analysis, minimizing variability in platform design, content structure, and instructional methods. Additionally, MOOCs, which have been extensively utilized before, during, and after the pandemic, offered high-quality educational content and served as a valuable resource for students in economically disadvantaged regions (Kennedy, 2014; Shao & Chen, 2021), where a significant portion of our participants resided.

To assess and compare students' experiences and perceptions of online learning, data were collected in China during three distinct time periods: 2016–2017, 2020, and 2024. The longitudinal design of the study is illustrated in Fig. 1. In China, the pandemic was first detected in December 2019 and peaked in February 2020. All the students were forced to study online during the pandemic. Thus, data gathered in 2020 reflected students' perceptions of online learning during the pandemic, while

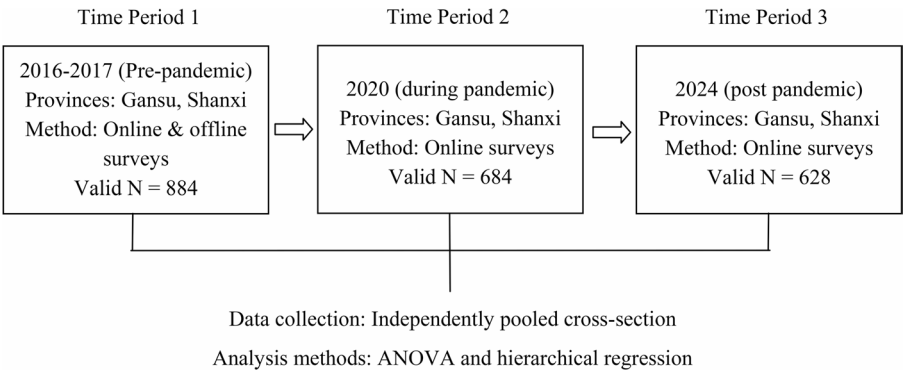

Fig. 1 Summary of the longitudinal design

data from 2016 to 2024 represented students' perceptions before and after the pandemic. The independently pooled cross-section across time method was utilized to collect the data. In addition, to make the results comparable, college students from two provinces (i.e., Gansu and Shanxi) in China were recruited in the samples. These two provinces are considered economically disadvantaged regions in China. As such, MOOCs have been actively promoted and widely adopted by universities to increase equity in educational opportunities and enhance access to educational opportunities for all students. The questionnaire was administered mainly through online channels (such as social networking sites and survey platforms). Finally, we gathered 884 valid questionnaires from 2016 to 2017, 684 valid questionnaires in 2020, and an additional 628 valid questionnaires in 2024. The demographic information of the respondents is reported in Table 1.

3.2 Validity and reliability

To test the discriminant validity of the constructs, exploratory factor analysis with Varimax rotation was employed. The results, presented in Table 2, show that the items for perceived usefulness, performance-to-cost value, and accessibility exhibited high loadings on the same factor, suggesting a strong conceptual overlap between them (Hair et al., 2019). Consequently, these three variables were integrated into a single construct, termed "perceived benefits", for the subsequent analysis. Similarly, the items for lack of publicity and lack of instruction were loaded together, leading to their combination into one construct, denoted as "lack of support". The items for the remaining variables remained separate and all the factor loadings were above 0.5.

Table 1 Sample demographics

Demographics	Category	Period 1 (Pre) (N=844)		Period 2 (During) (N=684)		Period 3 (Post) (N=628)	
		Count	Percent	Count	Percent	Count	Percent
Gender	Male	249	29.5%	191	27.9%	190	30.3%
	Female	595	70.5%	493	72.1%	438	69.7%
Age	18 and below	56	6.7%	38	5.6%	18	2.9%
	19–20	312	37.0%	353	51.6%	162	25.8%
	21–22	351	41.5%	216	31.6%	249	39.6%
	23–24	79	9.3%	29	4.3%	156	24.8%
	25 and above	21	2.5%	6	0.8%	42	6.7%
	Not answer	25	3.0%	42	6.1%	1	0.2%
Education level	Associate degree	33	3.9%	12	1.8%	33	5.3%
	Bachelor's	781	92.5%	660	96.5%	453	72.1%
	Master's	7	0.8%	1	0.1%	131	20.9%
	Doctorate	2	0.2%	0	0%	7	1.1%
	Others	25	2.6%	11	1.6%	4	0.6%
Online learning experience	Almost Never	224	26.5%	23	3.4%	59	9.4%
	Occasionally	319	37.8%	98	14.3%	196	31.2%
	Sometimes	195	23.1%	102	14.9%	181	28.8%
	Often	70	8.3%	137	20%	132	21%
	Very Often	36	4.3%	324	47.4%	60	9.6%

Table 2 Factor analysis of the barriers

Items	Factor Loading						Alpha
	1	2	3	4	5	6	
Useful3	**0.760**	−0.067	0.181	−0.028	−0.194	0.144	0.850
Value3	**0.734**	−0.144	0.027	−0.119	0.036	−0.032	
Value2	**0.700**	−0.114	−0.016	−0.033	0.065	−0.079	
Useful2	**0.685**	−0.050	0.244	−0.051	−0.200	0.234	
Useful1	**0.680**	−0.044	0.259	−0.049	−0.150	0.264	
Value1	**0.590**	−0.184	−0.007	0.050	0.155	−0.204	
Access2	**0.574**	−0.314	−0.123	0.058	0.071	−0.226	
Access1	**0.570**	−0.270	0.156	−0.053	0.061	−0.020	
Access3	**0.551**	−0.318	−0.036	−0.005	0.132	−0.177	
Public2	−0.168	**0.781**	−0.007	0.008	−0.043	−0.010	0.775
Public3	−0.064	**0.700**	−0.212	0.044	0.008	−0.257	
Instruct1	−0.254	**0.665**	−0.131	0.112	−0.179	0.065	
Instruct2	−0.321	**0.602**	−0.198	0.109	−0.104	0.008	
Public1	−0.230	**0.582**	−0.076	−0.081	0.074	−0.197	
Self-regulate2	0.081	−0.116	**0.854**	−0.037	0.006	−0.088	0.817
Self-regulate1	0.095	−0.117	**0.821**	0.007	−0.039	0.071	
Self-regulate3	0.142	−0.173	**0.795**	−0.064	0.029	−0.013	
Interact2	−0.051	0.038	−0.009	**0.847**	0.083	0.034	0.801
Interact3	−0.064	0.067	−0.039	**0.839**	0.099	0.045	
Interact1	−0.029	0.008	−0.042	**0.819**	0.063	−0.031	
Tradition2	0.034	−0.024	−0.019	0.088	**0.845**	0.141	0.751
Tradition1	0.066	−0.121	−0.059	0.118	**0.808**	0.048	
Tradition3	−0.087	−0.026	0.093	0.065	**0.644**	0.357	
Peers1	−0.079	−0.123	−0.007	0.042	0.276	**0.773**	0.735
Peers2	0.002	−0.120	−0.048	0.009	0.170	**0.753**	

The reliability of these constructs was tested using Cronbach's alpha, with all values exceeding 0.7, thereby meeting the requirements for reliability test (Hair et al., 2019).

3.3 Analysis methods

First, to examine changes in students' intention to engage in online learning over time, a one-way ANOVA followed by a post-hoc Tukey HSD test was conducted. Then to compare the influences of the proposed constructs on students' intentions to use online learning before, during, and after the pandemic, we analyzed the data using multiple regression with three-wave independently pooled cross-section samples. This method, employing randomly collected samples from the population at different points in time, allows for more precise estimators and more powerful test statistics (Wooldridge, 2016). In the analysis, the dependent variable was the intentions to use MOOCs in future studies. Two dummy variables, Y20 and Y24, representing the years 2020 and 2024 respectively, were created, with the earliest year in the sample (2016–2017) serving as the base year. By interacting the year dummy variables with key explanatory variables, we were able to determine whether the effect of each variable changed over time. The factors we control for include gender, age, education level, and usage experience. By examining the t statistic of the interaction term, we

Table 3 One-way ANOVA with post-hoc Tukey HSD test

The dependent variable is students' intention to use online learning

Source	Sum of Squares	df	Mean Square	F	*p*-value
Between Groups	8.938	2	4.469	10.449	<0.001
Within Groups	920.854	2153	0.428		
Total	929.792	2155			

Table 4 Post-hoc Tukey HSD comparisons

Year (I)	Year (J)	Mean Difference (I-J)	Standard Deviation	*p*-value
2020	2016–2017	−0.099	0.034	0.003
2024	2016–2017	0.064	0.034	0.063
2024	2020	0.163	0.036	<0.001

can test whether the changes in the variables' impacts on intentions are significant or not. The analysis was conducted by using SPSS 29.0.

4 Findings

The one-way ANOVA with post-hoc Tukey HSD test results are shown in Tables 3 and 4. One-way ANOVA results indicated significant differences in students' intention to use online learning over time ($F=10.449$, $p<0.001$) (Table 4). This indicates that students' intentions varied significantly over time, prompting further investigation using post-hoc tests to identify where these differences occurred. Specifically, post-hoc Tukey HSD tests revealed a significant decline in intention during the pandemic (2020) compared to the pre-pandemic period (2016–2017) (mean difference = −0.099, $p<0.01$). However, by 2024, students' intention to engage in online learning had significantly rebounded compared to 2020 (mean difference = 0.163, $p<0.001$), reflecting a notable post-pandemic recovery.

The regression analysis concerned with the effects of different constructs on intentions analysis results is presented in Table 5. All the independent variables included in the model accounted for 40.7% of the variance in predicting students' intentions to use MOOCs in their future studies (adjusted $R^2=0.407$, $F=60.653$, $p<0.001$). This indicates that the overall conceptual model is statistically significant. In the first block, respondents' gender, age, education level, and online learning usage experience were included as control variables. The analysis revealed that the coefficient for gender ($\beta=0.075$, $p<0.01$) was significantly associated with the intentions to use MOOCs. Since 'male' was coded as 0 and 'female' as 1, this indicates that female students are more likely to adopt MOOCs in their future studies. Additionally, the coefficient for education level ($\beta=0.042$, $p<0.1$) was significant at the 0.1 significance level (Hirschauer et al., 2022), suggesting that respondents with higher education degrees are inclined to use MOOCs in their future studies. Moreover, the positive association of usage experience ($\beta=0.061$, $p<0.001$) with intentions indicates respondents who frequently engage with MOOCs are more likely to continue using them.

Table 5 Regression analysis with the three-wave pooled cross sections

Independent variables	β	Standardized β	t test	p value
Gender	0.075	0.052	3.018	0.003
Age	−0.002	−0.006	−0.275	0.783
Education Level	0.042	0.034	1.723	0.085
Usage Experience	0.061	0.124	6.022	<0.001
Based Year (2016–2017) (Pre)				
Perceived Benefits	0.547	0.452	13.272	<0.001
Lack of Interactivity	0.036	0.042	1.218	0.223
Self-Regulation	0.131	0.154	4.720	<0.001
Learning Tradition	−0.089	−0.105	−3.148	0.002
Peers' impact	0.005	0.007	0.213	0.831
Lack of Support	−0.061	−0.064	−1.900	0.058
Year 2020 (During)				
Year20	0.672	0.474	1.609	0.108
Perceived Benefits*Year20	−0.189	−0.534	−3.000	0.003
Lack of Interactivity*Year20	−0.006	−0.017	−0.165	0.869
Self-Regulation*Year20	0.070	0.168	1.719	0.086
Learning Tradition*Year20	0.011	0.029	0.245	0.806
Peers' impact*Year20	0.000	0.001	0.007	0.994
Lack of Support*Year20	−0.195	−0.336	−3.626	<0.001
Year 2024 (Post)				
Year24	−0.270	−0.189	−0.705	0.481
Perceived Benefits*Year24	0.103	0.307	1.717	0.086
Lack of Interactivity*Year24	−0.006	−0.015	−0.148	0.882
Self-Regulation*Year24	0.025	0.058	0.664	0.507
Learning Tradition*Year24	−0.082	−0.206	−2.106	0.035
Peers' impact*Year24	0.014	0.032	0.435	0.663
Lack of Support*Year24	−0.100	−0.164	−2.161	0.031
F	60.653 ($p<0.001$)			
Adjusted R^2	40.7%			

"Year20" indicates the year 2020, and "Year24" denotes the year 2024

In the block of base year 2016–2017 (pre-pandemic), perceived benefits ($\beta=0.547$, $p<0.001$) were found to have a strong positive influence on students' intentions to use MOOCs, suggesting it was the strongest predictor of intention at the time. Self-regulation ($\beta=0.131$, $p<0.001$) also positively impacted students' intentions to use MOOCs, indicating that students with better self-management skills were more inclined to engage in MOOCs, though the influence was moderate in magnitude. In contrast, learning tradition ($\beta=-0.089$, $p<0.01$) and lack of support ($\beta=-0.061$, $p<0.1$) had negative, albeit weaker, effects, implying that traditional learning preferences and insufficient instructional support posed barriers to online learning adoption.

In the block of year 2020 (during the pandemic), the impact of perceived benefits ($\beta=-0.189$, $p<0.01$) on students' intentions to use MOOCs was lower than in 2016–2017 (i.e., the coefficient is $0.547-0.189=0.358$), indicating that the effects of perceived usefulness became weaker during the pandemic. On the contrary, the

impact of self-regulation ($\beta = 0.070$, $p < 0.1$) was higher in 2020 than in 2016–2017 (i.e., the coefficient is $0.131 + 0.070 = 0.201$), suggesting that independent learning skills became increasingly critical in self-paced learning environments. As for the effect of lack of support ($\beta = -0.195$, $p < 0.001$), it was found that its negative influence on intentions was significantly intensified during the pandemic in 2020 (i.e., the coefficient is $-0.061 - 0.195 = -0.256$), reflecting the barriers derived from insufficient support became more pronounced during the pandemic. However, the influences of the remaining variables did not show significant changes in 2020 compared to 2016–2017.

In the analysis for the year 2024 (post-pandemic), we found that the impact of perceived benefits ($\beta = 0.103$, $p < 0.1$) was higher than in 2016–2017 (i.e., the coefficient is $0.547 + 0.103 = 0.65$), suggesting a renewed appreciation for the value of online learning. The negative influence of learning tradition ($\beta = -0.082$, $p < 0.05$) on intentions was further exacerbated (i.e., the coefficient is $-0.089 - 0.082 = -0.171$) in 2024 after the pandemic compared to 2016–2017, indicating students' stronger preferences for traditional instruction. Similarly, the negative lack of support ($\beta = -0.100$, $p < 0.05$) in 2024 was increasingly magnified (the coefficient is $-0.061 - 0.100 = -0.161$) compared to 2016–2017, suggesting that post-pandemic online learning environments still require improved instructional and institutional support.

To better illustrate the evolving effects of the constructs, the fluctuations in their regression coefficients over time are presented and compared in Fig. 2. Firstly, the

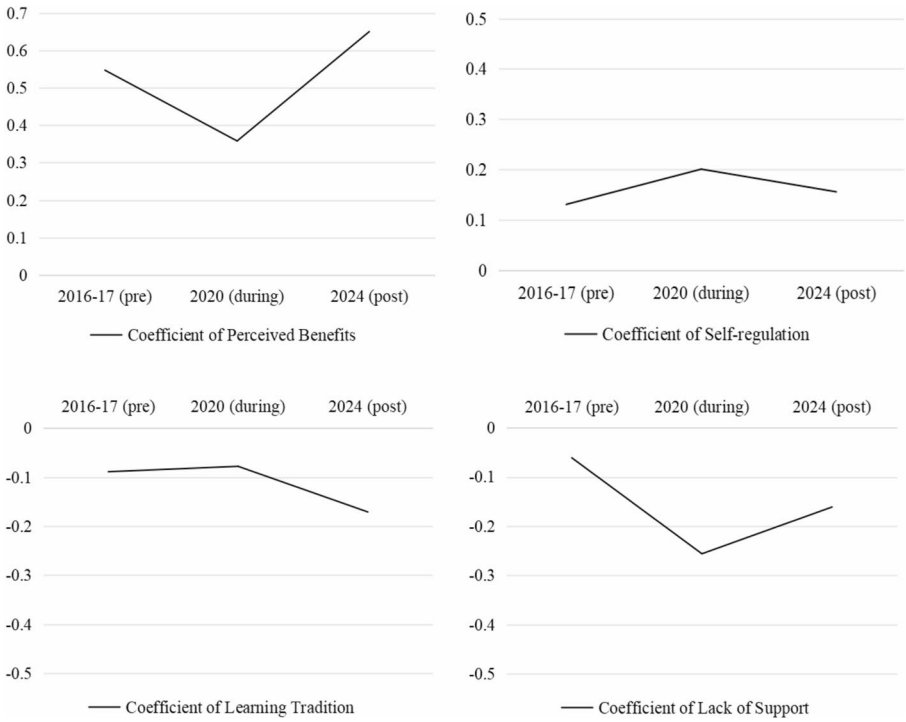

Fig. 2 Fluctuation of Construct Coefficients Over Time

findings reveal a sharp decline in the impact of perceived benefits on students' intentions to use online learning during the pandemic, followed by a notable rebound in the post-pandemic period. Second, the influence of self-regulation slightly increased during the pandemic compared to the pre-pandemic period. Third, the negative influence of learning tradition became more pronounced after the pandemic, suggesting a stronger resistance to online learning over time. Additionally, the negative effect of lack of support on students' intentions was significantly intensified during the pandemic and remained substantial in the post-pandemic phase. In contrast, the effects of other variables, such as lack of interactivity and peer influence, were not statistically significant across the three waves. The implications of the findings are discussed in the following section.

5 Discussion

Framing the pandemic as a disruption to learning, this study seeks to explore how students' online learning experiences before, during and after the pandemic have shaped their intentions to engage in online learning. It further examines how the influence of these factors has evolved over time. The results reveal a significant decline in students' intentions to engage in online learning during the pandemic, followed by a clear rebound in the post-pandemic period. Specifically, through regression analysis using three-wave independently pooled cross-section samples, we found that the positive impact of perceived benefits of online learning decreased during the pandemic but increased afterward. Conversely, the influence of self-regulation increased during the pandemic and decreased slightly after the pandemic. Regarding learning tradition, its negative influence remained consistent before and during the pandemic, but was significantly amplified after the pandemic. The negative influence of the lack of support experienced intensified during the pandemic and did not improve afterward. However, lack of interactivity and peers' impact were found to exert no significant influence on students' intentions to use online learning.

Firstly, the findings indicate a sharp decline in the impact of perceived benefits on students' intentions to use online learning during the pandemic. With online learning often being the sole available option during this period, the lack of social interactions and disrupted academic planning made it less appealing. These demotivating and unengaging experiences led students to either engage in superficial learning or drop out, especially among students in economically disadvantaged regions (Gherghel et al., 2023). Moreover, their achievement goals, engagement, and perceptions of success all diminished (Daniels et al., 2021), leading them to doubt their abilities and learning objectives (Usher et al., 2021). Therefore, it is unsurprising to see that the perceived benefits of online learning significantly decreased during the pandemic. Despite the predominantly negative online learning experiences during the pandemic, the advantages of online learning, such as its flexibility, accessibility, and the ability to customize learning experiences, remain evident. Expectedly, students recognized these merits post-pandemic, leading to an increased perception of the positive impact of online learning benefits post-pandemic. The findings highlight the importance of sustained efforts by policymakers and institutions to clearly communicate and dem-

onstrate the value of online learning in order to maintain student engagement and motivation.

In terms of self-regulation, it was found that the impact of self-regulation increased significantly during the pandemic and decreased slightly in the post-pandemic era, although the drop was not statistically significant. During the pandemic, students had to learn at home. Spending several hours in front of a computer without interactions made it challenging for students to stay motivated and focused on their online courses (Gherghel et al., 2023). Additionally, the home environment presented numerous distractions, including social media, and family interactions, which made it difficult for students to concentrate. This challenge was particularly pronounced among learners from disadvantaged regions, where home-based learning conditions were often far from conducive (Aguilera-Hermida, 2020; Lemay et al., 2021). They struggled with distractions and concentration underscored the importance of developing strong self-regulation skills and required students to exert greater effort to regulate their learning processes (Han et al., 2023; Luo & Zhou, 2024). Therefore, educational institutions should incorporate digital literacy and self-directed learning skills into their curricula, particularly for students transitioning to online or blended learning environments.

As for the impact of learning tradition, we found that its negative influence of learning tradition on students' intentions to use online learning was significantly amplified in the wake of the pandemic. This suggests that students are increasingly inclined to prioritize face-to-face classroom experiences over online formats, indicating a strong preference for traditional educational methods. This is understandable as students preferred classroom learning because of diminished motivation, insufficient communication and interaction, and feelings of isolation in online environments (Alawamleh et al., 2020). Importantly, the compromised course quality, coupled with students' negative experiences and learning fatigue during the pandemic, has led to entrenched attitudes toward online learning that may not easily change (Lemay et al., 2021). Another possible explanation lies in the timing of data collection, which occurred in 2020 when in-person interactions were still restricted across various contexts (e.g., business, social) in most parts of China due to the pandemic. This may have influenced participants' preference for in-person connections (e.g., traditional learning preferences). This challenge is significant and particularly difficult to address in the aftermath of the pandemic, especially in developing regions where resources and infrastructure may be limited. The situation is even more concerning given that these areas often lack the necessary support systems to effectively enhance online learning, making it crucial to prioritize targeted interventions that can help bridge the gaps and improve learning outcomes (Ma & Lee, 2019). To address this, hybrid models should be presented as enhancements rather than substitutes for traditional teaching, combining flexibility with in-person learning elements.

Furthermore, the lack of support (i.e., inadequate instruction and publicity) was found to have a more pronounced negative impact on students' intentions to use online learning during the pandemic. During the pandemic, higher education institutions were forced to swiftly transfer teaching and learning online. However, to create a meaningful online learning experience, it is not merely a simple transfer of traditional classroom methods to a digital format, but requires careful design, effective pedagogy, and appropriate infrastructure (Fujita, 2020; Shao et al., 2023). Due to the

lack of support from educational institutions, both instructors and students struggled to adapt to online learning settings (Baltà-Salvador et al., 2021; Shirish et al., 2021). These made students feel frustrated and stressed, resulting in superficial participation (Fang et al., 2023; Hussein et al., 2020). Although this negative influence has diminished since the pandemic, it remains significant and continues to have detrimental effects. The findings highlight the need for investment in comprehensive learner support systems, including instructor and student training, timely technical assistance, and accessible learning platforms.

This study found that interactivity did not significantly influence students' intentions to use online learning, either before or after the pandemic. One possible explanation is that the lack of interactivity has been a persistent and well-recognized limitation of online learning environments (Maqableh & Alia, 2021; Selvaraj et al., 2021). As a result, students have become accustomed to this limitation, leading to relatively low expectations for interactivity. Similarly, peers' impact was found to have no significant impact on students' intentions to use online learning across the three waves. Online learning inherently offers a high degree of independence and autonomy, allowing students to schedule their study time and pace freely (Ali et al., 2018; Chiu, 2021). In this self-directed learning environment, the influence of peers is relatively weak. These findings suggest that both interactivity and peer impact have remained consistently limited in shaping students' behavioral intentions toward online learning over time, highlighting the enduring autonomy and individualism embedded in online learning contexts.

6 Conclusion

This study advocates that online learning is essential for the future as it supports flexible, accessible, and lifelong education in an increasingly digital world. The findings have contributed to understanding online learning dynamics amidst and following the COVID-19 pandemic. Importantly, the findings allow educational institutions to provide better support for student engagement, improve online learning outcomes, and prepare for future disruptions, ensuring that online learning remains a viable and effective option in both traditional and crisis-driven contexts. Several key points deserve elaboration. Firstly, by identifying and analyzing the evolving impacts of technological, user, and environmental factors based on the TUE framework, this research extends the framework's applicability into a longitudinal perspective. Specifically, the study highlights how elements such as perceived benefits, self-regulation, learning traditions, and lack of support have dynamically shaped students' engagement. The longitudinal findings enhance our understanding of the evolving nature of student attitudes, shedding light on the complex interplay between technology adoption, individual behaviors, and socio-educational contexts within the broader environmental forces. This comprehensive perspective is crucial for addressing the challenges of online learning and fostering its long-term acceptance. Furthermore, the adoption of a longitudinal design utilizing three-wave pooled cross-section data significantly enhances the robustness and credibility of the findings. This approach allows for a more nuanced analysis of the changes in students' intentions over time,

providing a clearer understanding of evolving trends. By capturing data across pre-, during, and post-pandemic periods, this approach overcomes the limitations of studies that rely solely on cross-sectional data, offering deeper insights into the disruptive nature of the COVID-19 pandemic and how it shaped both short and long-term online learning experiences.

The findings of the study call for educators and educational institutions to take proactive measures to mitigate the adverse effects of the pandemic on students' attitudes towards online learning. As the perceived benefits of online learning have decreased following the pandemic, educational institutions can help shift students' perceptions, reduce resistance to online learning, and encourage broader acceptance of digital learning as a viable and effective alternative to traditional instruction. Economically disadvantaged regions were particularly hard hit by the pandemic as students had fewer resources to adapt to online learning, leading to poor online learning experiences and exacerbating existing inequalities. Policymakers in these regions must prioritize investment in digital infrastructure, equitable access to devices, and training for both students and teachers to ensure that future online learning experiences are equitable and empowering. Importantly, students should be made aware that their online learning experiences during the pandemic were atypical and do not represent future online learning environments. Additionally, to enhance online learning support, universities should provide comprehensive technical assistance to help students effectively navigate online platforms and promptly troubleshoot issues. They should also experiment with emerging and innovative technologies, such as artificial intelligence (AI), augmented reality (AR), and virtual reality (VR), to transform negative perceptions of online education into positive immersive experiences (Ratten, 2023). In response to students' continued preference for traditional learning, policymakers and leaders in higher education can advocate hybrid learning models that integrate the structured, interactive elements of face-to-face instruction with the adaptability and reach of digital platforms. This can ease the transition for students who are accustomed to traditional classroom settings. Last but not least, recognizing that self-regulation plays a crucial role in shaping students' intentions to engage in online learning, universities should provide targeted courses or modules dedicated to developing essential self-regulation skills, such as goal-setting, time management, and digital study strategies. Additionally, the development and dissemination of customized digital learning toolkits, including apps for tracking learning goals, scheduling study time, and conducting self-assessments, can help students monitor their progress and stay organized.

Several limitations of the study should be acknowledged. To mitigate the influence of confounding variables arising from diverse learning platforms, the survey specifically concentrated on the usage of MOOCs. This focus ensures a consistent online learning environment accessible to all participants, allowing for more accurate comparisons and analyses of the data. Nevertheless, online learning encompasses a broad spectrum of modalities, including virtual classrooms, blended learning, and webinars. Future extension could broaden the scope to include a broader range of online learning modalities to validate the findings of the study. Secondly, the survey was limited to two economically disadvantaged regions in China, which may impact the generalizability of the findings. Future research could be conducted in diverse geographical

regions to capture a more comprehensive understanding of students' experiences and perceptions. Another limitation of this study is the lack of a panel design, as the three waves of data were collected from independent cross-sectional samples rather than tracking the same individuals over time. Future research should consider employing longitudinal panel designs or latent growth modeling to confirm the stability and causality of these findings. Lastly, this study did not consider the impacts of additional contextual variables, such as school types (e.g., public vs. private, urban vs. rural). These factors could influence students' access to online learning resources and their overall engagement. Future studies should consider incorporating these contextual indicators to enable more nuanced subgroup analyses and enhance external validity. Despite these limitations, the findings of the present study contribute to a more nuanced understanding of the dynamic effects of the factors shaping students' intentions to use online learning throughout the pre-, during, and post-pandemic phases. They shed light on the key aspects that warrant focused attention to restore students' interests and motivations in online learning.

Table 6 Questionnaire

Construct	Items	Questions
Perceived Usefulness	Useful1	Help to solve problems
	Useful2	Help to solve problems
	Useful3	General usefulness
Performance-to-cost value	Value1	The cost of MOOCs is acceptable
	Value2	The performance-to-price value is high
	Value3	The usage of MOOCs is worthwhile
Lack of Interactivity	Interact1	Inconvenience to discuss with the lecturers
	Interact2	Difficult to interact with other learners online
	Interact3	Lack of channels to exchange ideas effectively
Accessibility	Access1	Internet access to MOOCs (e.g., through Wifi)
	Access2	Necessary devices to learn MOOCs
	Access3	Able to find suitable MOOC resources
Self-regulation	Self-regulate1	I am good at resisting distractions
	Self-regulate2	I am self-disciplined
	Self-regulate3	I am able to keep on the schedule planned
Learning Tradition	Tradition1	Preferences for traditional teaching methods with instructors
	Tradition2	Preference for tradition ways of learning
	Tradition3	Habits and routines prohibiting the usage of MOOCs
Peers' impact	Peers1	Easy to be impacted by the majority's opinions
	Peers2	Social influence from peers
Lack of Instruction	Instruct1	University's help to guide the usage of MOOCs (R)
	Instruct2	Availability of instructions to find suitable MOOCs (R)
Lack of Publicity	Public1	MOOCs' awareness from advertisement (R)
	Public3	MOOCs' awareness from school organization (R)
	Public3	MOOCs awareness from promotional campaigns (R)
Intention to use online learning	Intent1	Willing to learn MOOCs in the future study
	Intent2	Plan to learn MOOCs in the next six months
	Intent3	Will add the links of MOOCs that interest me to my favorites

Appendix

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Long Ma. The first draft of the manuscript was written by Long Ma. Chei Sian Lee commented and revised on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests None.

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Authors and Affiliations

Long Ma[1] · Chei Sian Lee[2]

✉ Long Ma
malong@zjgsu.edu.cn

Chei Sian Lee
leecs@ntu.edu.sg

[1] School of Business Administration, Modern Business Research Center, Zhejiang Gongshang University, Hangzhou, China

[2] Wee Kim Wee School of Communication and Information, Nanyang Technological University, Singapore, Singapore